Data quality check: Demo Store

Period: 2026-10-05 — 2026-10-12 · Checked on 2026-10-23 11:53 UTC

12

Score out of 100
The data is distorted — decisions based on it are risky

Critical	High	Medium	Low
5	26	10	1

Key findings

• **Duplicate transactions.** 35 transactions (3.3%) were recorded more than once; 35 extra rows. Duplicated value: 3,981.

• **Loss: Orders in GA4.** “ga4” contains only 78% of the records in “crm” (1040 of 1330). 290 are missing.

• **staging is losing data compared with raw.** Unique keys: 1,500 in “raw” vs 1,293 in “staging” (ratio 0.862).

• **Days not loaded into staging.** 1 day(s) present in “raw” are missing in “staging”: 2026-10-09.

• **mart is losing data compared with staging.** Metric: 141,330 in “staging” vs 130,957 in “mart” (ratio 0.927).

Value of records missing from the checked systems: Orders in GA4 — $34,358. Your reports and ad platform algorithms can't see these sales.

What changed since the last check

New issues

Severity	Issue	Details
High	Data collection gap	1 interval(s) with fewer than 35% of the usual events. About 167 events are missing.
Critical	Duplicate transactions	35 transactions (3.3%) were recorded more than once; 35 extra rows. Duplicated value: 3,981.
High	Personal data where it shouldn't be	Column “event_params” contains email. In an analytics or ad platform this breaks the platform's policies.
High	Loss concentrated in the “platform” slice	Only 71% found for “web”, versus 94% for “android”.
High	Traffic source lost: Orders in GA4	For 14% of records the source is known in “crm” but empty or direct in “ga4”.
High	Dip versus reference: Purchases by hour	In 2 intervals “ga4” sees less than half of its usual share (86%).
Critical	staging is losing data compared with raw	Unique keys: 1,500 in “raw” vs 1,293 in “staging” (ratio 0.862).
Critical	Days not loaded into staging	1 day(s) present in “raw” are missing in “staging”: 2026-10-09.
Medium	Day-level differences: raw → staging	On 2 day(s) values differ by more than 1%.
High	Duplicates introduced when loading into staging	“staging” has 25 more rows than unique keys (0 in “raw”). The load is not idempotent or a JOIN multiplies rows.

Severity	Issue	Details
High	Amounts differ: raw → staging	175,577 vs 154,778 (ratio 0.882).
High	Records lost between raw and staging	207 of 1500 keys from “raw” are missing in “staging”. Examples: 100115, 100225, 100308, 100410, 100638, 100692, 100713, 100821, 100822, 100857, 100858, 100859, 100860, 100861, 100862, 100863, 100864, 100865, 100866, 100867.
Critical	mart is losing data compared with staging	Metric: 141,330 in “staging” vs 130,957 in “mart” (ratio 0.927).
Medium	Day-level differences: staging → mart	On 5 day(s) values differ by more than 1%.
Medium	Sales dashboard: “Orders” doesn't match the data	The dashboard shows 1,132, the data 1,190 (-4.9%). Days differing by more than 2%: 2.

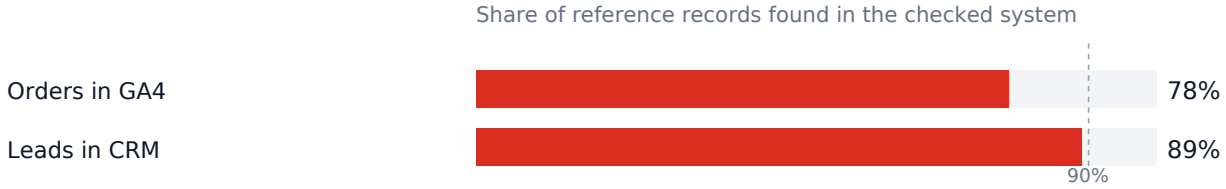
Resolved issues

Severity	Issue	Details
High	Amounts don't match: Orders in GA4	Amounts differ for 100.0% of matched records (median ratio 0.80). One source seems to exclude tax or shipping.

Metric changes

Metric	Before	After	Change
hubspot: records	573.0	534.0	-7%
Orders in GA4: match rate	0.9241	0.782	-15%
Leads in CRM: match rate	0.955	0.89	-7%
Purchases by hour: volume ratio	0.9457	0.8242	-13%
Orders raw → staging: volume ratio	0.9967	0.862	-14%
Revenue staging → mart: volume ratio	1.0	0.9266	-7%

Cross-system reconciliation



Reconciliation	Reference	Checked system	Found	Missing	Extra	Value lost
Orders in GA4	crm	ga4	78.2%	290	36	$34,358
Leads in CRM	ga4	hubspot	89.0%	66	0	—
Purchases by hour	crm	ga4	82.4% of volume	—	—	—
Orders raw → staging	raw	staging	86.2% of volume	—	—	—
Revenue staging → mart	staging	mart	92.7% of volume	—	—	—

Where records are lost: Orders in GA4

Slice	Value	Records	Found
platform	web	920	71%

Slice	Value	Records	Found
utm_source	google	689	77%
utm_source	yandex	266	77%
utm_source	(empty)	186	80%
utm_source	email	123	81%
utm_source	vk	66	89%
platform	ios	165	93%
platform	android	245	94%

Where records are lost: Leads in CRM

Slice	Value	Records	Found
utm_source	vk	102	87%
utm_source	yandex	94	88%
utm_source	(direct)	80	89%
platform	web	600	89%
path	/contact	600	89%
utm_source	google	201	89%
utm_source	email	123	91%

Warehouse: dwh

Table	Columns	Rows	Time column	First record	Last record
raw_orders	$5	$1,500	created_at	2026-10-05 00:00:58	2026-10-11 23:57:53
stg_orders	$5	$1,318	created_at	2026-10-05 00:00:58	2026-10-11 23:57:53
mart_daily_sales	$3	$6	day	2026-10-05	2026-10-11
raw_events_backup	$2	$0	created_at	None	None

Dashboards vs data

Dashboard	Metric	Day	On dashboard	In data	Difference
Sales dashboard	Revenue	2026-10-11	$17,229	$25,457	-32.3%
Sales dashboard	Revenue	2026-10-05	$19,158	$23,087	-17.0%
Sales dashboard	Revenue	2026-10-06	$18,926	$22,552	-16.1%
Sales dashboard	Orders	2026-10-06	$170	$200	-15.0%
Sales dashboard	Orders	2026-10-05	$159	$187	-15.0%
Sales dashboard	Revenue	2026-10-10	$25,854	$26,809	-3.6%
Sales dashboard	Revenue	2026-10-08	$22,033	$22,267	-1.1%
Sales dashboard	Revenue	2026-10-07	$21,036	$21,157	-0.6%
Sales dashboard	Orders	2026-10-08	$189	$189	+0.0%
Sales dashboard	Orders	2026-10-07	$201	$201	+0.0%
Sales dashboard	Orders	2026-10-10	$210	$210	+0.0%
Sales dashboard	Orders	2026-10-11	$203	$203	+0.0%

Website: what tags send

Page	gtm	ga4	xtracker	dataLayer
Home page	container_loaded	page_view×2	page_view	—
Checkout	container_loaded	page_view	page_view	purchase
About	—	—	—	—
Click “Top up”	container_loaded	page_view×2, deposit_click	page_view, element_click	deposit_click

What to do

1. Duplicate transactions (critical)

Send a unique transaction_id and don't resend the purchase when the thank-you page is reopened (a cookie/localStorage flag, or send from the server). Check that two sending methods aren't active at the same time.

2. Loss: Orders in GA4 (critical)

Find the step where the event gets lost: compare the list of missing records with your tag firing logs. If the loss is concentrated on the website, check whether the tag has time to load on the thank-you page; sending purchases from the server (Measurement Protocol / server-side GTM) is more reliable.

3. staging is losing data compared with raw (critical)

Data is lost between layers: compare the filters and JOINs in the transformation and trace the records from the “keys not loaded” list.

4. Days not loaded into staging (critical)

Whole days didn't reach the next layer — re-run the load for those dates and find out why they were skipped.

5. Paid orders reach GA4 (at least 90%) (high)

A business rule agreed with you is violated. Details are in the “Business rules” section.

6. Sales dashboard: “Revenue” doesn't match the data (high)

The dashboard number doesn't match the data: check the dashboard's filters, report time zone and metric formula.

7. Duplicates introduced when loading into staging (high)

Duplicates appear during the load: make it idempotent (MERGE by key) and check JOINs for row multiplication.

8. page_view is sent 2 times: Home page (high)

page_view is sent more than once: keep a single GA4 tag (GTM or code) and check History Change triggers.

9. Paid click recorded as direct (high)

Keep gclid / fbclid and UTM parameters across the session and pass them with the conversion event. Check auto-tagging and referral exclusions.

10. Payment provider shown as a traffic source (high)

Add the payment provider's domain to the referral exclusion list (GA4: “List unwanted referrals”).

11. Traffic source lost: Orders in GA4 (high)

Add hidden UTM / click-ID fields to your forms and check that the CRM saves them when a lead is created. Check forms inside iframes and form clones in A/B tests.

12. Personal data where it shouldn't be (high)

Remove personal data from URLs and event parameters (parameter filtering in GTM, clean-up on the website). Google's policies and privacy laws require it.

13. Personal data is sent to analytics: Checkout (high)

The browser sends personal data to analytics: remove it from URLs and event parameters.

14. Data collection gap (high)

Compare the time of the gap with website releases and GTM changes — the tag most likely wasn't working during that period.

15. Loss concentrated in the “platform” slice (high)

The loss is uneven: start fixing with the worst slice (device, page or source) named in the issue.

16. No analytics: About (high)

The tracking code doesn't work on this page: check that the GTM container or tracker is installed on this page template and isn't blocked.

17. Scenario “Click “Top up””: deposit_click was not sent to xtracker (high)

The event exists on the website, but no tag sends it: add a tag/trigger in GTM or a rule in the tracker.

18. Event “deposit_click” leaves the website but is missing in “ga4” (high)

The event leaves the website but never reaches the data: check property filters, the export and the warehouse load.

19. Tags fire before cookie consent: Home page (high)

Tags fire before cookie consent: set up Consent Mode or consent-based triggers.

20. Empty table: raw_events_backup (high)

The table is empty: check that the load into it runs at all and writes to the right schema.

21. Time offset between sources: Orders in GA4 (medium)

Use one time zone in all systems (we recommend UTC in the warehouse) — otherwise daily reports disagree.

22. Extra records: Orders in GA4 (medium)

Review the extra records: they are test or cancelled orders, or the reference export is incomplete.

23. Consent button not found: About (low)

Check the consent button selector in the site audit settings.

Business rules

Rule	Result	Explanation
Paid orders reach GA4 (at least 90%)	**Failed**	Found 1040 of 1330 (78.2%); required: at least 0.9.
Order ID is unique in GA4	**Failed**	Duplicate “txn_id” keys: 35.
CRM leads have a source (at least 85%)	**Failed**	358 of 534 (67.0%); required: at least 0.85.
No paid test orders under $1	**Failed**	18 rows violate the rule.
CRM export updated in the last 30 hours	**Passed**	Last record 4 h ago; allowed: 30 h.

All issues found

Severity	Category	System	Issue and details
Critical	Duplicates	ga4	**Duplicate transactions** 35 transactions (3.3%) were recorded more than once; 35 extra rows. Duplicated value: 3,981.
Critical	Completeness	ga4	**Loss: Orders in GA4** “ga4” contains only 78% of the records in “crm” (1040 of 1330). 290 are missing.
Critical	Warehouse	staging	**staging is losing data compared with raw** Unique keys: 1,500 in “raw” vs 1,293 in “staging” (ratio 0.862).
Critical	Warehouse	staging	**Days not loaded into staging** 1 day(s) present in “raw” are missing in “staging”: 2026-10-09.
Critical	Warehouse	mart	**mart is losing data compared with staging** Metric: 141,330 in “staging” vs 130,957 in “mart” (ratio 0.927).
High	Business rules	ga4	**Paid orders reach GA4 (at least 90%)** Found 1040 of 1330 (78.2%); required: at least 0.9.
High	Business rules	ga4	**Order ID is unique in GA4** Duplicate “txn_id” keys: 35.
High	Business rules	hubspot	**CRM leads have a source (at least 85%)** 358 of 534 (67.0%); required: at least 0.85.
High	Dashboards	Sales dashboard	**Sales dashboard: “Revenue” doesn't match the data** The dashboard shows 124,236, the data 141,330 (-12.1%). Days differing by more than 2%: 4.
High	Duplicates	staging	**Duplicates introduced when loading into staging** “staging” has 25 more rows than unique keys (0 in “raw”). The load is not idempotent or a JOIN multiplies rows.
High	Duplicates	website	**page_view is sent 2 times: Home page** GA4 received several page_view hits for a single page load — pageviews and sessions are inflated. Usually this means two GA4 tags (in code and in GTM) or a history-change trigger.
High	Duplicates	website	**page_view is sent 2 times: Click “Top up”** GA4 received several page_view hits for a single page load — pageviews and sessions are inflated. Usually this means two GA4 tags (in code and in GTM) or a history-change trigger.
High	Traffic sources	ga4	**Paid click recorded as direct** 131 conversions came with an ad click ID (gclid — 131), but the source is empty or direct. Ad platforms are under-credited.
High	Traffic sources	ga4	**Payment provider shown as a traffic source** 31 conversions are credited to the payment provider (checkout.stripe.com). Add it to the referral exclusion list.
High	Traffic sources	ga4	**Traffic source lost: Orders in GA4** For 14% of records the source is known in “crm” but empty or direct in “ga4”.
High	Traffic sources	hubspot	**Traffic source lost: Leads in CRM** For 33% of records the source is known in “ga4” but empty or direct in “hubspot”.
High	Personal data	ga4	**Personal data where it shouldn't be** Column “event_params” contains email. In an analytics or ad platform this breaks the platform's policies.
High	Personal data	website	**Personal data is sent to analytics: Checkout** Found in requests to ga4: email.
High	Completeness	ga4	**Data collection gap** 1 interval(s) with fewer than 35% of the usual events. About 167 events are missing.

Severity	Category	System	Issue and details
High	Completeness	ga4	**Loss concentrated in the “platform” slice** Only 71% found for “web”, versus 94% for “android”.
High	Completeness	hubspot	**Loss: Leads in CRM** “hubspot” contains only 89% of the records in “ga4” (534 of 600). 66 are missing.
High	Completeness	ga4	**Dip versus reference: Purchases by hour** In 2 intervals “ga4” sees less than half of its usual share (86%).
High	Website	website	**No analytics: About** Not a single request to GA4, Yandex Metrica, Meta or a tracker left this page.
High	Event collection	website	**Scenario “Click “Top up””: deposit_click was not sent to xtracker** After the scenario steps, xtracker did not receive “deposit_click”.
High	Event collection	ga4	**Event “deposit_click” leaves the website but is missing in “ga4”** The browser sent “deposit_click” to ga4, but source “ga4” has no such event for the whole period — it is lost on the way to storage or filtered out.
High	Consent	website	**Tags fire before cookie consent: Home page** 3 requests were sent before the consent button was clicked (ga4, xtracker).
High	Consent	website	**Tags fire before cookie consent: Checkout** 2 requests were sent before the consent button was clicked (ga4, xtracker).
High	Consent	website	**Tags fire before cookie consent: Click “Top up”** 3 requests were sent before the consent button was clicked (ga4, xtracker).
High	Amounts	staging	**Amounts differ: raw → staging** 175,577 vs 154,778 (ratio 0.882).
High	Warehouse	dwh	**Empty table: raw_events_backup** The table has no rows.
High	Warehouse	staging	**Records lost between raw and staging** 207 of 1500 keys from “raw” are missing in “staging”. Examples: 100115, 100225, 100308, 100410, 100638, 100692, 100713, 100821, 100822, 100857, 100858, 100859, 100860, 100861, 100862, 100863, 100864, 100865, 100866, 100867.
Medium	Business rules	crm	**No paid test orders under $1** 18 rows violate the rule.
Medium	Time	ga4	**Time offset between sources: Orders in GA4** The same record appears in “ga4” on average +4.0 h later than in “crm”. Likely different time zones — daily reports will disagree.
Medium	Time	ga4	**Hourly offset: Purchases by hour** The series align best with a -4 h shift (correlation 0.75 vs -0.15). The sources probably use different time zones.
Medium	Dashboards	Sales dashboard	**Sales dashboard: “Orders” doesn't match the data** The dashboard shows 1,132, the data 1,190 (-4.9%). Days differing by more than 2%: 2.
Medium	Traffic sources	ga4	**Conversions without a source** 26% of orders/leads have no source or show direct / (not set).
Medium	Traffic sources	hubspot	**Conversions without a source** 42% of orders/leads have no source or show direct / (not set).
Medium	Completeness	ga4	**Extra records: Orders in GA4** 36 records exist in “ga4” but not in “crm” — test or cancelled orders, or an incomplete reference export.

Severity	Category	System	Issue and details
Medium	Event collection	website	**dataLayer event “purchase” was not sent anywhere: Checkout** The website pushed the event, but neither GA4 nor the tracker sent it — a tag or trigger is missing.
Medium	Warehouse	staging	**Day-level differences: raw → staging** On 2 day(s) values differ by more than 1%.
Medium	Warehouse	mart	**Day-level differences: staging → mart** On 5 day(s) values differ by more than 1%.
Low	Consent	website	**Consent button not found: About** No element “#accept” on the page (or the banner didn't appear). Check the selector; a banner inside an iframe can't be clicked.

Data checked

Source	System	Records	Period in data
crm	generic	1 467	2026-10-04 20:00 — 2026-10-11 19:57
ga4	ga4_bigquery	7 573	—
hubspot	hubspot	534	2026-10-05 00:13 — 2026-10-11 23:38

Personal data is not carried into the report or normalized tables: emails and phone numbers are replaced with hashes.